12

Lecture 12

Mobility Management Entity (MME)

موجودیت مدیریت تحرک (MME)

2G — 3G — **4G** — 5G — 6G

COURSE PART
Part II – 4G access and core (Lectures 5–13)

DIFFICULTY
Intermediate

ESTIMATED STUDY TIME
6–8 hours (3 h reading, 3–5 h exercises)

PREREQUISITES
Lectures 4, 6, 9 and 11; basic queueing (M/M/1)

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Original lecture notes for the course *Advanced Communication Networks*. The lecture sequence follows the course syllabus (26 lectures, 2G to 6G). Equations use identical notation in the English and Persian editions. Companion edition: Persian (`lecture-12-fa.pdf`).

1 Learning objectives

After studying this lecture you should be able to:

- **LO1** **List** the functions and reference points of the MME and **explain** why it is a pure control-plane node.
- **LO2** **Decode** a GUTI and **explain** how temporary identities protect privacy and route signalling inside an MME pool.
- **LO3** **Trace** the initial attach procedure end to end.
- **LO4** **Evaluate** paging strategies by their expected signalling cost and delay.
- **LO5** **Dimension** an MME pool with weight factors and failure margins, and **analyse** MME overload with a queueing model.
- **LO6** **Compare** the MME with the 2G/3G SGSN/VLR and the 5G AMF.

2 Prerequisite knowledge

You need the EPC architecture and EMM/ECM states of Lecture 9, the S1AP and RRC procedures of Lecture 6, location-area design from Lecture 4 and the S6a dialogue with the HSS (Lectures 10–11). The overload model uses the M/M/1 queue: Poisson arrivals at rate λ, exponential service at rate μ, one server.

Key terms. The vocabulary of this lecture with its Persian equivalents:

Term	Persian	Meaning in this lecture
MME	موجودیت مدیریت تحرک	Control-plane node of the EPC that terminates NAS signalling.
NAS	لایهٔ غیردسترسی	Signalling between UE and MME (EMM and ESM) [7].
GUTI	شناسهٔ موقت یکتای سراسری	Temporary UE identity allocated by the MME.
S-TMSI	TMSI کوتاه‌شده	MMEC + M-TMSI; used in paging and RRC set-up.
TAI list	فهرست شناسه‌های ناحیهٔ ردیابی	Set of tracking areas in which the UE need not report.
Paging	فراخوانی	Locating an idle UE for mobile-terminated traffic.
MME pool	MME مخزن	Group of MMEs serving the same pool area (S1-flex).
Relative MME capacity	MME ظرفیت نسبی	Weight used by eNBs to share new UEs among pool members.
Overload control	کنترل اضافه‌بار	Mechanisms that reject or delay signalling when the MME is saturated.
T3412	زمان‌سنج TAU دوره‌ای	Periodic tracking-area update timer.

3 Introduction and motivation

In GPRS and UMTS the SGSN did everything for packet subscribers: it authenticated them, tracked them, held their sessions and forwarded their packets. The EPS split this node into two. The *Mobility Management Entity* (MME) keeps the control functions; the serving gateway keeps the packets [6]. The split lets operators scale signalling capacity (which grows with the number of devices and their mobility) independently of user-plane capacity (which grows with traffic volume), and it is the first step of the control/user-plane separation that 5G completes.

The MME is the UE's counterpart in the core: every NAS message – attach, tracking-area update, service request, bearer request – terminates there. This lecture looks at what the MME stores, how it locates idle devices, how MMEs share load in a pool, and what happens when signalling exceeds capacity.

4 Fundamental concepts

4.1 Functions of the MME

- **NAS signalling and security**: EMM and ESM procedures, authentication with vectors from the HSS, NAS ciphering and integrity [2], [7].
- **Reachability and mobility**: tracking-area lists, paging of idle UEs, inter-eNB handover with MME relocation (S10), inter-RAT mobility to 2G/3G (S3).
- **Session management**: default and dedicated bearer set-up via the S-GW (S11), selection of S-GW and P-GW.
- **Subscription handling**: registration with the HSS (S6a), enforcement of access restrictions, equipment-identity check (S13).
- **Interworking**: CS fallback and SMS over SGs [3], SRVCC over Sv [9], location services (SLs).

The MME never touches user packets; its load is measured in procedures per second and in stored contexts, not in bits per second.

Table 1. MME reference points.

Interface	Peer	Protocol	Purpose
S1-MME	eNB	S1AP over SCTP	UE-associated and non-UE signalling, NAS transport [5]
S11	S-GW	GTPv2-C	Session and bearer management [1]
S6a	HSS	Diameter	Authentication data, location update [4]
S10	MME	GTPv2-C	Context transfer on MME relocation
S3	SGSN	GTPv2-C	Inter-RAT mobility with 2G/3G
S13	EIR	Diameter	Equipment identity (IMEI) check
SGs	MSC/VLR	SGsAP over SCTP	CS fallback, SMS [3]
Sv	MSC	GTPv2-C	SRVCC handover of voice to CS [9]

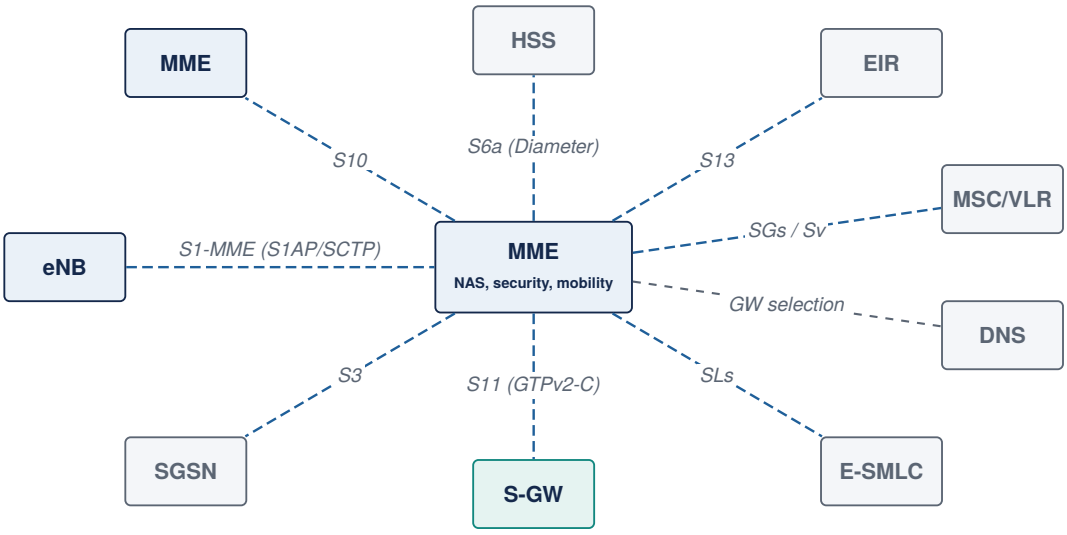


Figure 1. The MME and its reference points. Dashed lines are control-plane interfaces.

4.2 Identities

The permanent IMSI is sent over the air as rarely as possible. After the first attach the MME allocates a *Globally Unique Temporary Identity* (GUTI) [8] (Figure 2). Its first part, the GUMMEI,

names the MME that allocated it (PLMN, 16-bit MME group ID, 8-bit MME code); its last part, the 32-bit M-TMSI, names the UE within that MME. The 40-bit S-TMSI (MME code + M-TMSI) is enough inside one pool and is used in paging and RRC connection set-up. Reallocating the GUTI regularly prevents an observer from linking a UE's transactions.

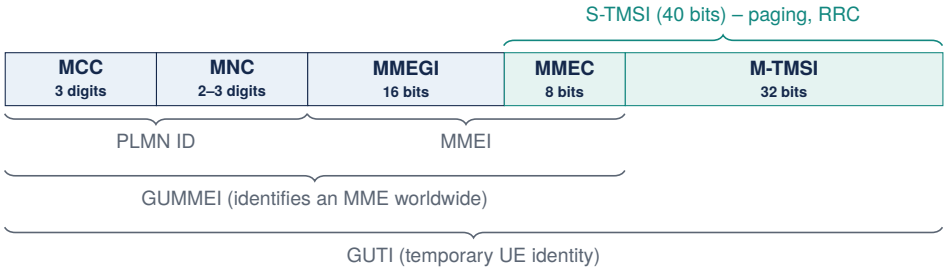


Figure 2. Structure of the GUTI and the identities derived from it.

4.3 MME pools

An eNB can connect to several MMEs (S1-flex). The MMEs that serve the same set of tracking areas form a *pool*. When a UE first connects, the eNB's NAS node selection function chooses an MME, weighted by the *relative MME capacity* each MME announced in S1 Setup; afterwards the MME code in the S-TMSI routes every new connection of that UE to the same MME. Pools provide load sharing, resilience (a failed MME's UEs re-register with the others) and the ability to take an MME out of service gracefully.

5 Detailed technical explanation

5.1 UE context

For each registered UE the MME stores the IMSI and GUTI, the security context (keys, NAS counts, algorithms), the subscription profile from the HSS, the TAI list and last known TAI and eNB, the EMM/ECM state, the selected S-GW and the bearer contexts (EBI, QoS, S-GW TEIDs). In ECM-CONNECTED the context also holds the S1AP identifiers of the UE's eNB connection. This state is what makes the MME the *anchor* for the UE's signalling: losing it (e.g. after an MME failure) forces the UE to attach again.

5.2 Tracking areas and paging

The MME gives each UE a list of tracking areas (TAI list); the UE performs a TAU only when it enters a TA outside the list or when the periodic timer T3412 expires (default 54 min) [7]. When downlink data arrive for an idle UE, the S-GW sends Downlink Data Notification and the MME pages the UE in the cells of its TAI list with its S-TMSI. Operators reduce paging load with *staged* strategies – first the last-seen eNB, then the last TA, then the whole list – at the price of extra delay (Section 6.1).

If no periodic TAU arrives, the MME starts the *mobile reachable timer* (slightly longer than T3412); when it expires, the MME stops paging the UE, and after a further implicit-detach timer it detaches it.

5.3 Gateway selection

At attach, the MME selects an S-GW that serves the UE's tracking area and a P-GW that serves the requested APN, normally by DNS queries on names built from the TAI and the APN, which return candidate gateways with priorities and weights. The P-GW must stay the same for the life of the PDN connection (it owns the IP address); the S-GW may change with mobility (Lecture 13).

5.4 Load balancing, rebalancing and overload control

Under normal conditions the eNBs distribute new UEs in proportion to the relative capacities. To empty an MME (for maintenance) or to rebalance, the MME releases UEs with the cause *load-balancing TAU required*: the UE performs a TAU, and because its S-TMSI is withheld, the eNB selects a new MME. When an MME is overloaded it sends S1AP Overload Start to its eNBs, which then reject new RRC connections of selected types (e.g. all except emergency and mobile-terminated) until Overload Stop [5]. NAS back-off timers and access-class barring (Lecture 8) complement this at the UE.

6 Mathematical models

6.1 Staged paging

Model. The MME pages in stages $s = 1, 2, \ldots$; stage s covers n_s cells and finds the UE with (unconditional) probability p_s, with $\sum_s p_s = 1$. Each stage takes one paging response time T_{p}. The expected number of cells paged and the expected delay are

$$\mathbb{E}[M] = \sum_{s} n_s \Big(1 - \sum_{j<s} p_j\Big), \qquad \mathbb{E}[D] = T_{\mathrm{p}} \sum_{s} \Big(1 - \sum_{j<s} p_j\Big). \tag{1}$$

Example. Three stages: last eNB ($n_1 = 1$, $p_1 = 0.6$), last TA ($n_2 = 20$, $p_2 = 0.3$), TAI list ($n_3 = 60$, $p_3 = 0.1$): $\mathbb{E}[M] = 1 + 0.4 \times 20 + 0.1 \times 60 = 15$ cells, compared with 60 for blanket paging; $\mathbb{E}[D] = 1.5\, T_{\mathrm{p}}$ instead of T_{p}.

Interpretation. Staging pays off when the probabilities are concentrated in the early stages (static users, small cells), and is unattractive for delay-sensitive services such as incoming voice calls, for which operators often page the whole list at once.

6.2 Pool weights and failure margin

Model. The share of new UEs that MME i receives is

$$s_i = \frac{w_i}{\sum_j w_j}, \tag{2}$$

with w_i its relative capacity. If a pool of m equal MMEs must survive the loss of one without overload, each MME may be loaded at most to

$$\rho_{\max} = \frac{m-1}{m}. \tag{3}$$

Example. Weights 100, 100, 50 give shares 0.4, 0.4, 0.2. A pool of four equal MMEs may run at 75 % in normal operation; a pool of two at only 50 %.

6.3 Processing delay and overload

Model. Treat the MME's signalling processor as an M/M/1 queue with capacity μ procedures per second and offered load λ ($\rho = \lambda/\mu < 1$). The mean response time is

$$W = \frac{1}{\mu - \lambda} = \frac{1}{\mu(1-\rho)}. \tag{4}$$

Example. $\mu = 5000\,\mathrm{s}^{-1}$: $W = 1$ ms at $\lambda = 4000$, 10 ms at 4900 and 100 ms at 4990 (Figure 3).

Interpretation. The delay grows without bound as $\rho \to 1$. Once response times approach NAS timers (T3410 = 15 s for attach), UEs retransmit, the offered load rises further and the system can collapse; overload control must therefore act well before $\rho = 1$. The M/M/1 model is optimistic for the bursty arrivals of a signalling storm.

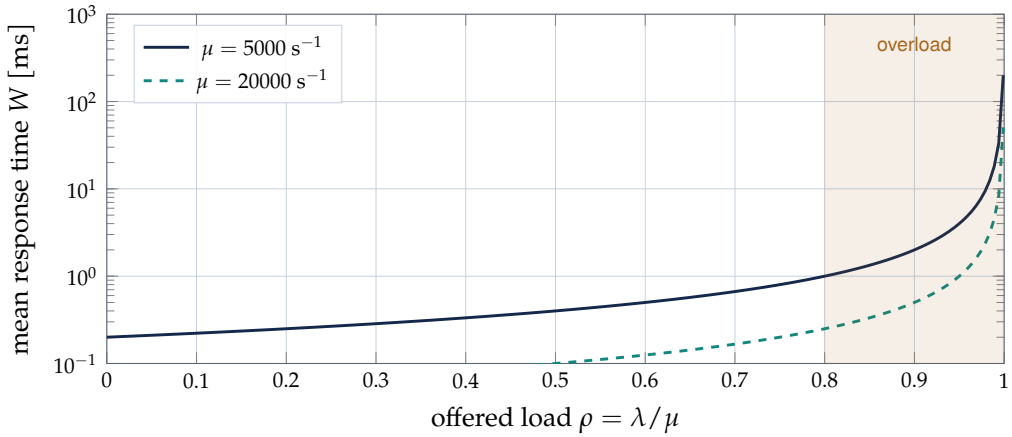


Figure 3. Mean response time of an M/M/1 signalling processor versus load (Eq. (4)).

7 Network architecture and diagrams

In a typical network, the eNBs of a region connect to all MMEs of a pool of two to eight MMEs; pools of neighbouring regions overlap so that a UE moving between them changes MME only at the pool boundary. The MMEs reach the HSS via DRAs, the S-GWs and P-GWs over S11 (often in the same data centre), and the MSC/VLR over SGs for CS fallback and SMS (Figure 1).

8 Step-by-step procedures

Procedure 1 • Initial attach (simplified)

1. The UE sends Attach Request with a PDN Connectivity Request inside RRC set-up; the eNB selects an MME and forwards it in S1AP Initial UE Message.
2. The MME identifies the UE (GUTI or IMSI), runs EPS-AKA with a vector from the HSS and activates NAS security.

3. The MME registers with the HSS (ULR/ULA) and receives the subscription profile.
4. The MME selects S-GW and P-GW and sends Create Session Request; the P-GW allocates the IP address and, with a PCRF, obtains the policy; responses return the S-GW's uplink tunnel endpoint.
5. The MME sends Initial Context Setup Request with Attach Accept and the default bearer to the eNB, which configures the radio bearer with RRC Connection Reconfiguration.
6. The UE answers Attach Complete; the MME sends Modify Bearer Request with the eNB's downlink tunnel endpoint to the S-GW. The default bearer now carries traffic in both directions.

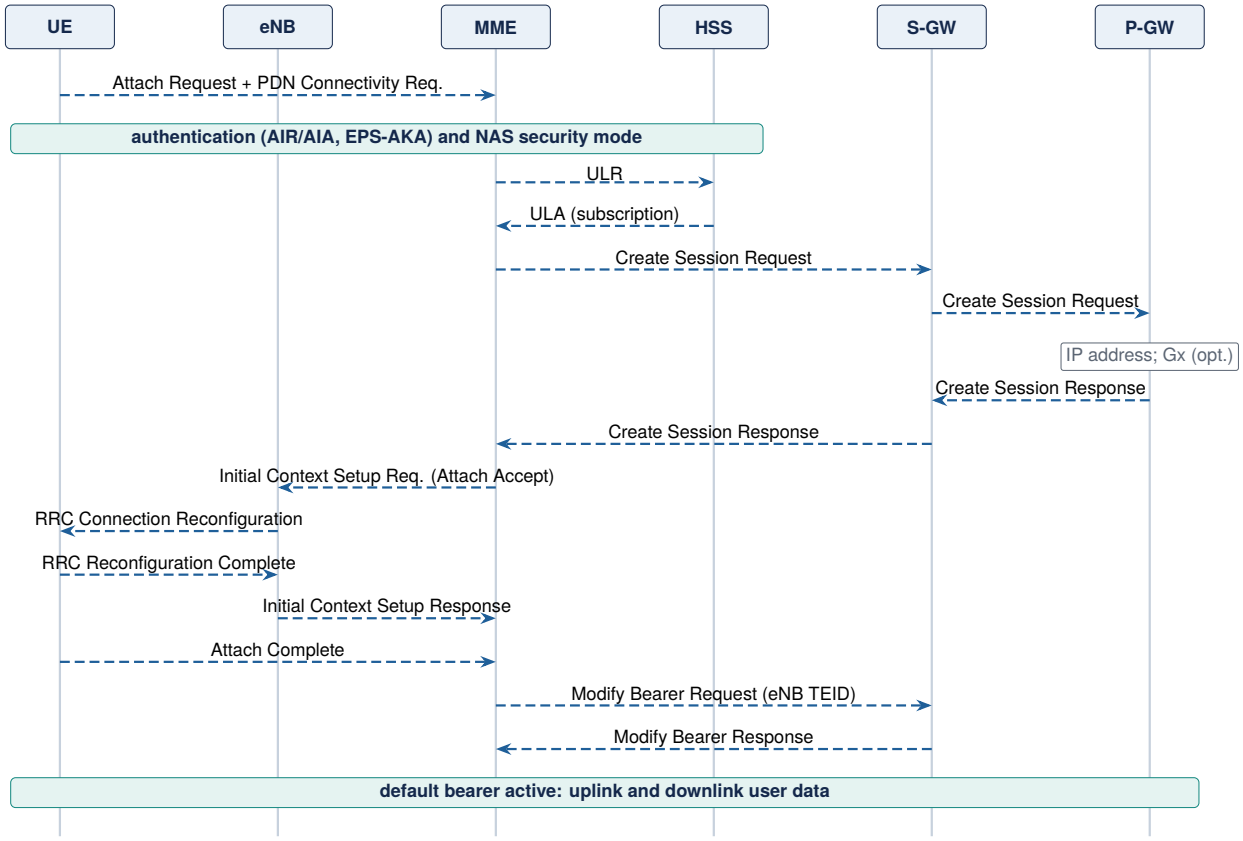


Figure 4. Simplified E-UTRAN initial attach (Procedure 1) [6].

Procedure 2 • Emptying an MME for maintenance

1. Set the MME's relative capacity to 0 (S1AP MME Configuration Update) so that eNBs stop selecting it for new UEs.
2. Release connected UEs with cause *load-balancing TAU required*; page idle UEs or wait for their next contact and release them in the same way.
3. Each UE performs a TAU without S-TMSI routing; the eNB selects another MME, which fetches the context over S10 and registers with the HSS.
4. When no contexts remain, the MME can be taken out of service.

9 Worked examples

Worked example 1 • Decoding a GUTI

Problem. A GUTI has MCC 432, MNC 35, MMEGI `0x8001`, MMEC `0x02` and M-TMSI `0xC0A80001`. Which identities does the eNB use, and which MME serves the UE?
Solution. GUMMEI = (432, 35, 32769, 2): MME code 2 in group 32769 of PLMN 432-35. The S-TMSI is `0x02C0A80001` (40 bits); paging and RRC set-up carry only this. The eNB's NAS node selection function maps MMEC 2 to the corresponding S1 connection in its pool.

Worked example 2 • Paging cost

Problem. A city has 5×10^5 mobile-terminated events per busy hour. Compare the paging messages per busy hour for blanket paging over 60 cells with the three-stage strategy of Section 6.1.
Solution. Blanket: $5 \times 10^5 \times 60 = 3.0 \times 10^7$ cell pagings per hour. Staged: $5 \times 10^5 \times 15 = 7.5 \times 10^6$, a factor 4 less, at a mean delay of $1.5\,T_{\mathrm{p}}$.

Worked example 3 • Pool sizing

Problem. A region has 6×10^6 registered UEs. Each MME supports 2×10^6 contexts. How many MMEs are needed if the pool must survive one failure?
Solution. Without redundancy $\lceil 6/2 \rceil = 3$. With one failure tolerated, the remaining $m-1$ MMEs must hold all contexts: $(m-1) \times 2 \times 10^6 \geq 6 \times 10^6 \Rightarrow m = 4$, each loaded to $6/8 = 75\,\%$ in normal operation, consistent with Eq. (3).

10 Technology comparison

Table 2. Mobility-management nodes across generations.

	SGSN + MSC/VLR (2G/3G)	MME (4G)	AMF (+ SMF) (5G)
User plane	SGSN forwards packets (3G: optional direct tunnel)	None	None
Session control	SGSN	MME (via S-GW)	SMF (separate)
Location unit	Location / routing area	TAI list	Registration area (TAI list)
Temporary ID	TMSI / P-TMSI	GUTI, S-TMSI	5G-GUTI, 5G-S-TMSI
Pooling	Iu-flex (optional)	S1-flex	AMF sets, NRF
Interfaces	MAP, Gn/Gp, Iu	S1AP, GTPv2-C, Diameter	NGAP, HTTP/2 SBI

From MME to AMF. The 5G AMF keeps the MME's access and mobility functions but hands session management to a separate SMF, and replaces GTPv2-C and Diameter with service-based interfaces (Lecture 19). The concepts – temporary identities, registration areas, pools, overload control – carry over almost unchanged.

11 Key takeaways

Key takeaways

- The MME is the EPC's control-plane anchor: NAS, security, mobility, bearer control, gateway selection; no user data.
- GUTI = GUMMEI + M-TMSI; the S-TMSI routes signalling to the right MME in a pool and protects the IMSI.
- Attach combines authentication, HSS registration, session creation and radio-bearer set-up.
- Staged paging trades delay for signalling: $\mathbb{E}[M] = \sum n_s(1 - \sum_{j<s} p_j)$.
- Pools share load by weights; $N+1$ redundancy limits normal load to $(m-1)/m$.
- Response time explodes as $\rho \to 1$; overload control must act early.

12 Exercises

Exercise 1 • Level 1 – conceptual • Control or user plane?

For each function, state whether the MME, the S-GW or the P-GW performs it: (a) paging; (b) buffering downlink packets for an idle UE; (c) IP address allocation; (d) NAS ciphering; (e) choosing the S-GW.

Exercise 2 • Level 1 – conceptual • Why temporary identities?

Give two reasons why the network uses the GUTI/S-TMSI instead of the IMSI in paging and RRC set-up.

Exercise 3 • Level 2 – numerical • Paging strategy

Evaluate Eq. (1) for $p = (0.4, 0.4, 0.2)$ and $n = (1, 30, 90)$. Compare with blanket paging over 90 cells. Which strategy would you use for incoming voice calls?

Exercise 4 • Level 2 – numerical • Weights after a failure

A pool has five MMEs with relative capacities $2, 2, 1, 1, 1$ (capacity proportional to weight). Compute the shares of new UEs before and after the failure of one weight-2 MME, and the maximum normal-operation utilisation that still allows the pool to absorb this failure.

Exercise 5 • Level 3 – analytical • Overload

An MME has $\mu = 5000$ procedures/s. (a) What load keeps $W \leq 10$ ms in the M/M/1 model? (b) After a regional power outage, 2×10^6 UEs try to attach within 60 s. Compare the offered load with μ and explain which mechanisms keep the system stable.

Exercise 6 • Level 3 – analytical • Choosing T3412

An MME holds 5×10^6 idle UEs. Compute the periodic-TAU rate for T3412 = 54 min and 12 min, and discuss the trade-off between signalling, UE battery and the time needed to detect unreachable UEs. Why do IoT devices use much longer values?

Solutions

Solution to Exercise 1

(a) MME (triggered by the S-GW's Downlink Data Notification); (b) S-GW; (c) P-GW; (d) MME; (e) MME.

Solution to Exercise 2

Privacy: the IMSI is permanent, so sending it in the clear would let an eavesdropper track a subscriber; the GUTI is reallocated. Efficiency and routing: the 40-bit S-TMSI is shorter than the IMSI and its MME code tells the eNB which pool member holds the context.

Solution to Exercise 3

$\mathbb{E}[M] = 1 + 0.6 \times 30 + 0.2 \times 90 = 37$ cells versus 90; $\mathbb{E}[D] = (1 + 0.6 + 0.2)\, T_\mathrm{p} = 1.8\, T_\mathrm{p}$. For voice, call set-up delay is critical and the UE must be found at once: page the whole TAI list (or at least skip the single-cell stage).

Solution to Exercise 4

Before: $2/7 = 28.6\,\%$ for each weight-2 MME and $1/7 = 14.3\,\%$ for each weight-1 MME. After: total weight 5, so 40 % for the remaining weight-2 MME and 20 % for each weight-1 MME. Capacity falls from 7 to 5 units, so the normal utilisation must not exceed $5/7 = 71\,\%$.

Solution to Exercise 5

(a) $1/(\mu - \lambda) \le 0.01 \Rightarrow \lambda \le 4900\,\mathrm{s}^{-1}$ ($\rho \le 0.98$); with bursty traffic the real limit is lower. (b) $2 \times 10^6/60 = 3.3 \times 10^4$ attaches/s, almost seven times μ (and each attach involves several messages). Stability requires S1AP overload control (eNBs reject or delay connections), access-class barring and RACH back-off at the UE, randomised NAS retry timers, and a pool so that several MMEs share the load.

Solution to Exercise 6

$5 \times 10^6/3240 = 1543\,\mathrm{TAU/s}$ for 54 min; $5 \times 10^6/720 = 6944\,\mathrm{TAU/s}$ for 12 min. A short timer detects lost UEs sooner (detection takes about T3412 plus the reachability margin) but multiplies signalling and drains batteries. IoT devices send rarely and must last years on a battery; extended timers of many hours, combined with power saving mode, reduce their signalling and energy use, and the network accepts that they are unreachable most of the time.

13 Further reading

The MME's role and procedures are specified in TS 23.401 [6]; NAS in TS 24.301 [7]; S1AP in TS 36.413 [5]; GTPv2-C in TS 29.274 [1]; identities in TS 23.003 [8]. CS fallback and SRVCC are in TS 23.272 [3] and TS 23.216 [9]. Olsson et al. [11] and Sesia et al. [12] give textbook treatments; Kleinrock [10] covers the M/M/1 queue.

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